

Final report for MSc Project

**Temperature regulation system design for tracking changes in brain
organoid spontaneous activity over maturation**

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Abstract

This project primarily focuses on investigating the influence of temperature on the calcium dynamics of mature brain organoids. The organoid samples originate from human pluripotent stem cells. To facilitate this research, a feedback control system for temperature manipulation during imaging has been designed and tested. This system effectively maintains the organoid model at a stable temperature around normal body temperature.

The monitoring of temperature's impact on neural activity in the organoid model is significant as it validates the development of the model and its comparability of other research models. The organoid model has intrinsic benefit on tracking brain development and understanding underlying mechanism. The anticipation of future temperature validation is boosted by the present results from other adjustments made to organoid technology which aim to replicate the early brain developmental conditions.

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1. Introduction

The study of brain organoids emerged from the need to understanding the human brain development and fundamental mechanisms from a different view. Animal models are often inadequate for studying human brain disorders and drug testing due to morphological differences and functional complexity (Lancaster et al., 2013). Although the current development of brain organoids is limited by the challenges in understanding and duplicating the complicated cues and environment in early brain developmental stages, many evidences in morphological aspects, oscillatory waves and other stereotyped characteristics (Del Dosso et al., 2020) supporting the maturation of the organoid model and the compatibility to human brain development and function (Trujillo et al., 2019).

In this project, brain organoid samples are derived from human pluripotent stem cells (hPSCs) and generated with the protocol from Lancaster's group (Giandomenico et al., 2021). The classical methods in cell culture such as air-liquid interface (ALI) culture, the use of Matrigel for supplying the cell growth are applied in organoid culturing (Giandomenico et al., 2021). The three-dimensional cerebral organoids are plated down for long-term culture through the maturation process. Organoids samples are studied under a two-photon (2P) microscope with GCaMP staining labels the calcium ions within cells, especially in neurons, for monitoring real-time activity dynamics.

Organoid technology holds great potential to transform the field of neurological disease research and drug screening. Ethical considerations in organoid research predominantly arise from the utilisation of human stem cells and the exploration of certain potential applications such as the transplantation of the in vitro human organoid in animal models and gene engineering ("The Ethics of Brain Organoids | Eurostemcell"). However, organoid technology provides a chance to reduce the need for animal experimentation in preclinical testing, which at the same time, works as an accurate and personalised model for human brain development and neurodevelopmental disease study (Bredenoord et al., 2017). The in vitro organoid is more "transparent" for studying, for instance, compared to the preparation and staining in vivo, the live investigation of organoid samples with fluorescence imaging is much easier and valid for long-term tracking.

Regarding the topic of temperature analysis in the organoid model focused in this research, it is well aware that brain thermoregulation in homeothermic species, especially in human is critical. The temperature is tightly regulated and there is little understanding in the fundamental mechanism, regulation process and the impact of temperature fluctuation. The complexity and the ethical issue of the human model limited the chance to have an in-depth view of the spatiotemporal distribution of brain temperature, as well as the fluctuations by physiological and pathological factors under the mechanism of brain thermoregulation (Wang et al., 2014), since every action potential generation would generate heat and temperature fluctuation would affect the neural activity by impacting on the nerve conduction velocity, synaptic transmission and metabolic rate in reverse. The fundamental understanding of temperature dynamics in the brain could potentially be inspiring to explain the brain neuronal viability and impact on neurophysiological properties associated with brain disorders. Thus, brain organoids are a good alternative for live investigation.

Most research on brain organoids focused on the improvement of the culture condition, which has been proven with positive outcomes on organoid assembly. Engineering and mechanical solutions for regulating the nutrient fluid show significant influence on the organoid structure, which would also impact on functional development and mutation (Suong et al., 2021). However, temperature as a highly regulated factor in the human brain has not commonly been manipulated either within the organoid growing process (kept in an incubator) or during recording or imaging. It is meaningful to first validate the impact of temperature in the

matured organoid model by regulating the organoid sample temperature around the actual brain temperature during imaging, and further the development of organoid model with temperature regulation that mimics the functional brain.

2. Methods

This temperature control system works with an imaging chamber under 2P microscope (Fig.1). The organoid sample needs to be kept in a sterilised and sealed condition, for protecting the organoid from contamination and preventing any bubble formation which would disrupt the imaging view. So, the temperature sensor is not able to be directly positioned around the sample membrane or inside its medium for the same reason. Sensors are then fixed on the side surface of the chamber, with wires soldered to a connector. Temperature mapping is required between the sensor location and the internal temperature under the sealed condition before using for temperature monitoring. The mapping reliability is crucial for the later performance of temperature control under 2P microscopy.

For the purpose of using this system in a 2P imaging facility, within a tight operation area, the control circuit is soldered to an Arduino prototype shield and the temperature monitoring and the power supply modulation of the heating device are achieved by the Arduino (Uno R3) programming. Thermoelectric device (TED), or Peltier plate is powered from the main socket, with an 12V adapter. The chamber is placed on top of the Peltier plate, with a heatsink below. The sensor wires, Peltier plate and power adaptor are all connected to the main circuit via connectors. The chamber is compatible in the autoclave environment, under high operation pressure and temperature, the sensor connections are shielded and secured for standing frequent sterilisation process.

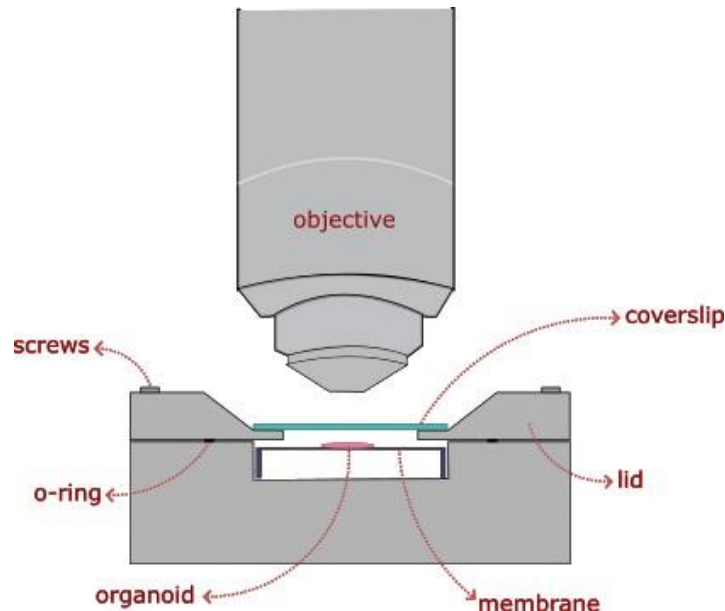


Figure 1: Side view of the imaging chamber, under the microscope objective lens

2.1 Chamber design

The chamber was specially designed for the “fragile” organoid sample and an sterilised and sealed inner cavity was prepared for the organoid with an observation window opened at the top, through the coverslip (Fig.1). The coverslip sealing with medium filled up between the top of the membrane to the coverslip. The O-ring, with the mechanical force of the screws, together created a leak-proof seal between the two metal parts.

The dimensions of this inner space is important for the mobility of the membrane, whereas a floating membrane could lead to significant drifting during imaging. Additionally, the working distance (around 3mm with the 20x objective lens used) of the microscope objective lens left a limited space in design from the organoid to the top of the coverslip. The opening on the lid was cut with a slope, which provided more free adjustment space for the objective lens.

Lastly, the slots cut for the temperature sensors, they were located on the side of base of the chamber. Three slots have been cut. The sensor slots were firstly filled with thermal paste (Electrolube TBS20S) to conduct the heat from main chamber body to the sensor, and then the sensors were hold in place with the glue (Araldite epoxy adhesive). Araldite mounted the sensors which could stand the autoclave environment and it is harden over time.

2.2 Temperature feedback control system

Adjustable organoid temperature during calcium imaging process is one of the main objectives of the study. An accurate, reliable and real-time reading is required for heating circuit control. The general considerations include the overshoot problem which might lead by the delay in the control response. The biological tissue would be in danger of permanently damage or denaturation in short time when cook in high temperature. The whole system should be ease to fit in the microscope shelter so each part is separable and pluggable to the main soldered board for arrangement. The heat dissipation of the Peltier module is complicated (Dziurdzia, 2011), plus the input signal oscillation caused by the connections, interference and temperature variation. This system is aiming to linearise the relationship between the control parameter calculated from the sensor reading and the heat output of the heating element, by controlling a pulse width modulated signal generated by Arduino.

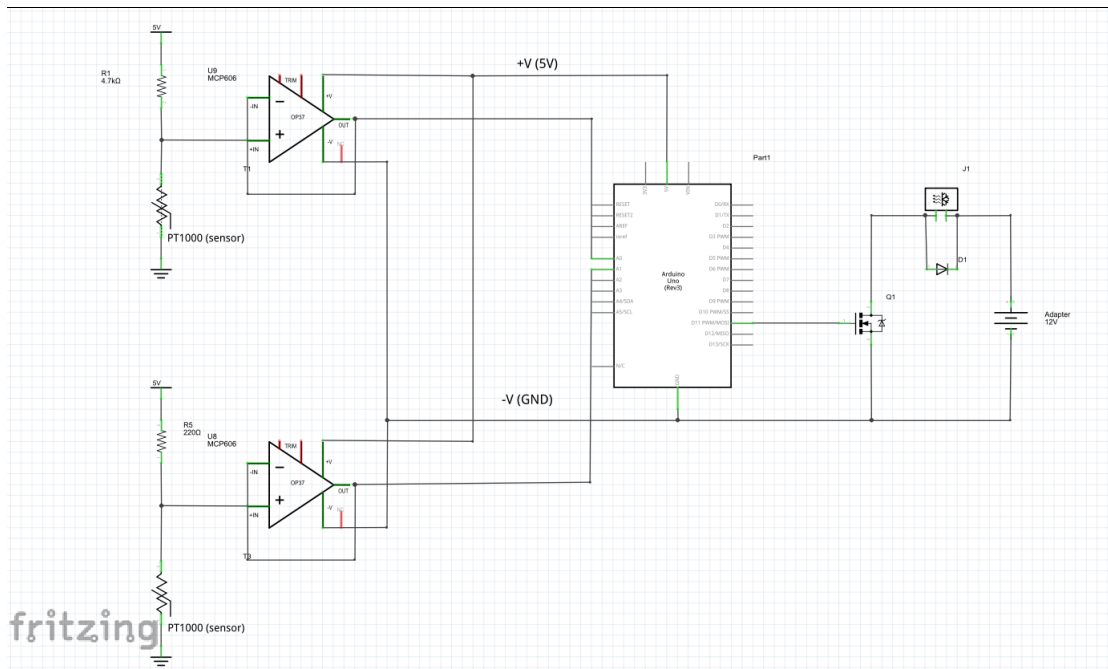


Figure 2: Monitoring sensor circuit (left) of the Arduino, and the heating circuit on the right side. The op amps and sensor circuits are supplied by Arduino 5V power, and the heating circuit by 12V adaptor.

The system has two main circuits, a monitoring circuit and a heating circuit (Fig.2). These two circuits are both connected and communicated to the microcontroller, Arduino. The monitoring circuit built two voltage followers with the isolated sensor readings as the output signal. A fix resistor (4.7kΩ) and the sensor (PT1000) was connected in series. Operational amplifier (op amp) OPA344 was chosen as it has rail-to-rail input and output with a stable unity gain, which delivered a reliable analog signal to the microcontroller. The voltage signals are then converted to digital reading through Arduino ADC and control the PWM at one of the set output pin of the Arduino. The detailed programming script is provided in the control algorithm section. In the heating circuit, a metal-oxide-semiconductor field-effect transistor (MOSFET, IRLB8314PbF) was used to control the current flow of the heating circuit. The field effect which the electrical conductivity of the semiconductor changes with the voltage applied to the Gate terminal of the MOSFET. Here, the microcontroller output signal is applied to the Gate terminal which switch on and off the current between its Source and Drain terminal, so across the heating module (Fig.2). A diode (BZX79 series) was added in parallel to the TED, to regulate voltage and protect the Peltier module from any reverse voltage during operation. A 12V power adapter was selected for powering the heating circuit, which provide a sufficient power rate under control.

The optimal temperature of cerebral brain to function is normally around 36.5°C to 37.5°C, in a range of 1-2°C. Therefore, a precise and stable temperature control is enquired for further validation of the temperature impact in the brain organoid model. This is achieved by several aspects in the system, for instance, the power sufficiency, sensor reading resolution, reading accuracy, temperature mapping accuracy and the feedback control algorithm.

TED performance test

The output power is primarily dependent on the power supply of the heating system since the microcontroller PWM adjusts the duty cycle to modulate the average power delivered. By testing under different clamp voltage, with same current limit (3A), provided by a bench DC power supply, the Peltier module was solely connected and the temperature of chamber placed on top of the module was recorded in time to investigate the TED heat dissipation rate (Dziurdzia, 2011), especially the rate of reaching the operating temperature range from the room temperature.

Temperature sensor resolution and accuracy

PT1000, made with Platinum which has resistance of 1000Ω at 0°C and rises around 4Ω with every 1°C increase. This sensor has a great linearity of its resistance to temperature relationship. The sensor was calibrated with a 0.1°C resolution, $\pm 0.2^\circ\text{C}$ accuracy thermometer (TME MM2000). The resolution of the sensor reading in the system was not only constraint by the calibration tool, but also strictly limited by the 10 bits value of ADC. To make use of a reasonable range of the 10 bits value mapped for 5V, the sensor signal was pre-processed before entering the Arduino (Fig.3). By removing the offset (Eq.1) and amplifying the subtracted signal (Eq.2), the amplitude of the input signal was able to be increased by 13 fold.

$$V_{out} = \frac{R_8}{R_{12}}(V_2 - V_1) \quad (1)$$

In Eq.1, V_1 is the offset voltage and V_2 is the sensor reading, where $\frac{R_8}{R_{12}}$ sets to 1 for unity gain.

$$A = \frac{V_{out}}{V_{in}} = 1 + \frac{R_{15}}{R_{14}} \quad (2)$$

In Eq.2, A is the amplification factor for the input and output signal, this is set by the choice of resistance ratio of the two resistor R_{15} and R_{14} .

Another attempt of improving the resolution of the reading was to change the default ‘analogReference’ to ‘INTERNAL’, which is an in-built reference of 1.1V in Uno board. This would theoretically improve the resolution in 4 times.

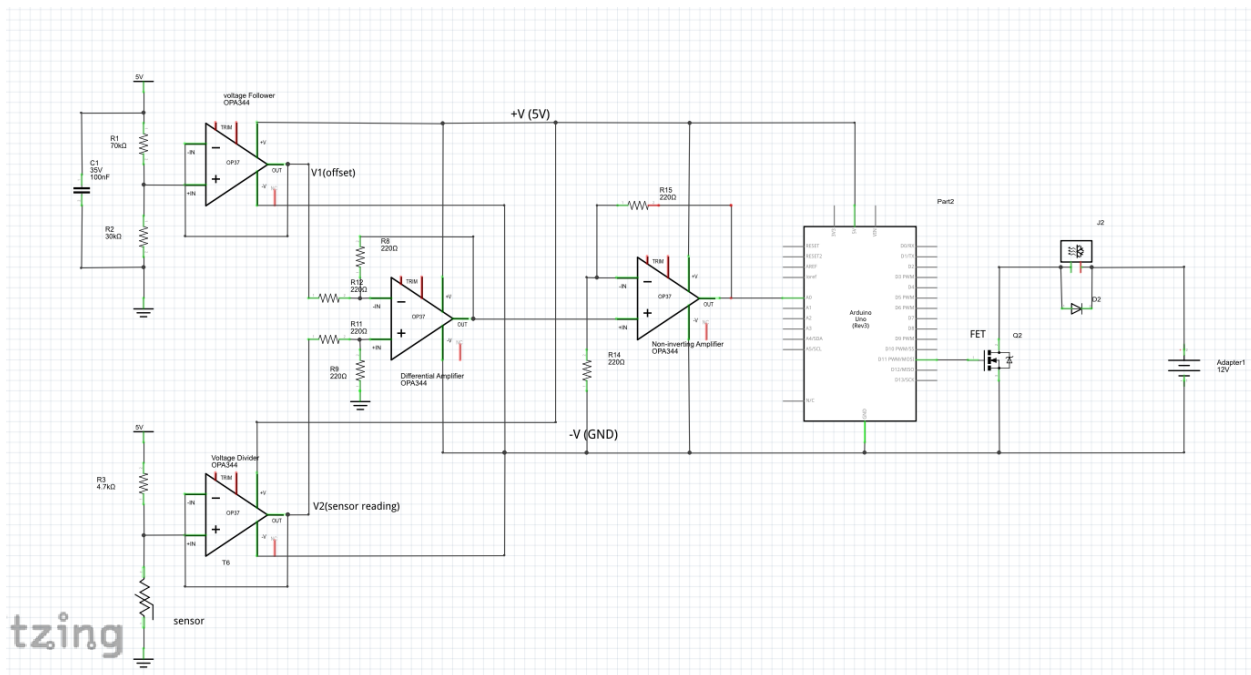


Figure 3: (left top op amp) Voltage follower, output an offset voltage; (left bottom) voltage follower, output one sensor reading; (middle) differential amplifier which subtract the offset from the sensor reading; then the signal went through a non-inverting amplifier before entering the Arduino Analog pin. Unchanged heating circuit on the right hand side of the microcontroller.

The temperature mapping between the sensor location and the sample location is crucial for the temperature control performance. The mapping was carried out under the ambient temperature in 2P facility with the circuit in Fig.2. A K-type thermocouple connected to TME thermometer was inserted and screwed in the chamber, with water filled up in the sample space. The sensor reading and thermometer readout was recorded synchronically, through the heating and cooling process (turning off the heating circuit power supply after stabilisation time). The system was run with setpoint and left on for 15 minutes to test the stability of the temperature control.

2.3 Control Algorithm

The system was operated with negative feedback loop (Fig.4), by subtracting the real-time reading with the desired setpoint and minimising the error. The controller here only included proportional control by setting up a proportional gain for the error (Eq.3).

$$\text{Output} = \text{gain} * \text{error} \quad (3)$$

The calculated output was then applied to the output pin of the microcontroller, which modulated the PWM duty cycle and therefore control the ratio of the “ON” state time (Fig.5).

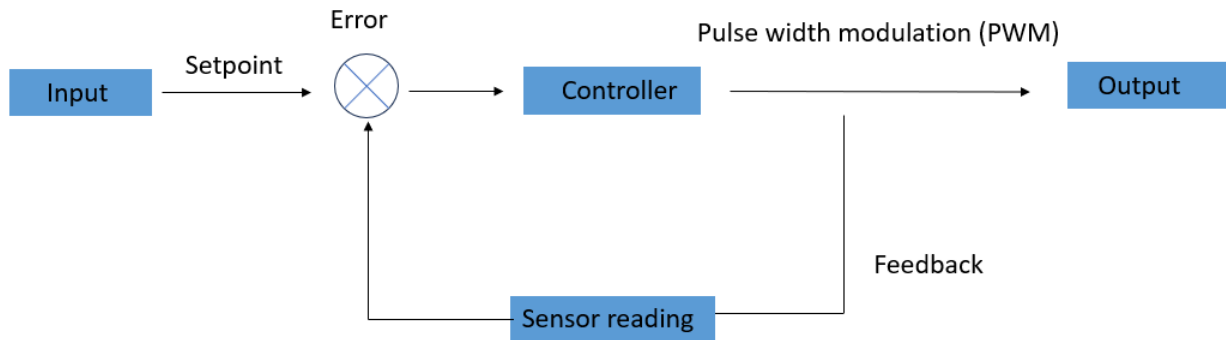


Figure 4: Feedback control loop

The control range was chosen previously in the power rate test, the maximum output value was set to 50 in the “analogWrite” syntax value, which is equivalent to 20% power “ON” as maximum. A deadband was included in the code, which delayed the activation of the power to prevent excessive control and fluctuations around the setpoint. The setpoint, a sensor reading value reference, was set from the temperature mapping result, and another parameter “gain” was a constant value that chose from a range of trials. The magnitude of the gain manipulated the temperature controlling range around the set temperature, a large gain value refers to a small control range and steep power reduction. For the best control outcome, two inserted sensor were mapped individually and averaged error was utilised for control algorithm (Eq.4). The performance was analysed by generating and recording a step response with setpoint adjustments.

$$\text{Error} = \frac{(\text{setpoint 1} - \text{sensor 1}) + (\text{setpoint 2} - \text{sensor 2})}{2} \quad (4)$$

```
Setpoint = mapped sensor value;
Gain = constant;
n = reading array size;
i=i++ ;
if i>=n{
In time window, average sensor readings}
Error = setpoint – averaged value;
Output = error * gain;
If (output <= 1*gain)
% deadband for delay in activation
PWM = 0
Else if (output>max desired capacity)
PWM = max capacity
Else
PWM=output
Zero stored array and averaged value
```

Figure 5: programming script outline, 100 readings were grouped for the average calculation, and the update rate of output was around 1Hz.

2.4 Organoid specimen preparation

The preparation was carried out in a cell culture hood, with the chamber and tools autoclaved. The organoid membrane was placed in the inner space of the chamber. Firstly, fill up the space beneath the membrane with the imaging medium and screwed up the lid. Then carefully add extra medium on top of the membrane till no air would left when place a coverslip on top. Organoids are easy to fall off the place as they are plated on top of the membrane and fragile for any shock. The preparation work is important for the imaging quality, as any sample movement and air bubble would disrupt the signal detection and cause artifacts. To enable the objective lens to scan through a wider distance range, the slope opening is mostly filled up, and distilled water is used to minimise the contamination (Fig.6).

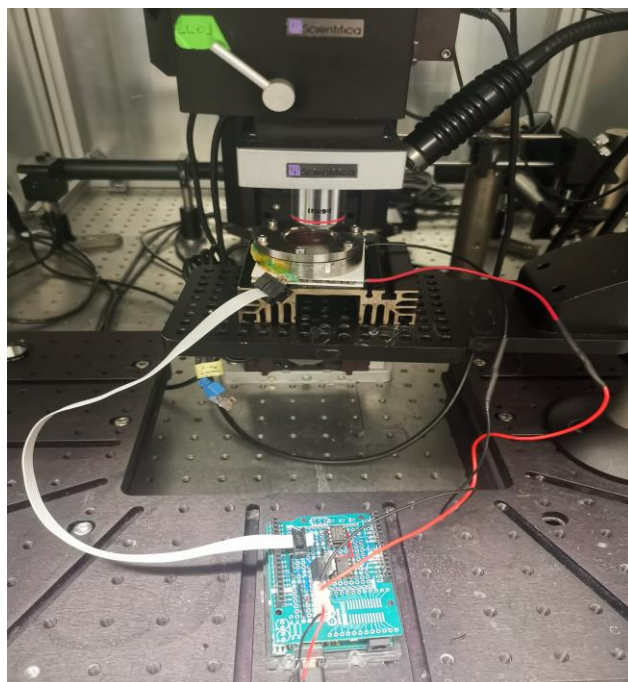


Figure 6: Temperature control system setup under the 2p microscope

3. Results

A 4V clamping voltage provided sufficient power to heat up the chamber from room temperature to the desired temperature range (35°C to 38°C) in a reasonable speed, which was under 2 minutes. However, considering that the commercial adaptor usually has a limited current and other integrated circuit for output regulation, a 5V, maximum current 3A adaptor was tested with the prototype and was not able to generate enough output power for heating up the chamber. 12V, 3A adaptor was chose instead and demonstrated a fast and controllable heating process.

The control algorithm used to provide the following results used one sensor reading for error calculation and the internal reference of ADC was applied for resolution improvement.

The temperature mapping of the sensor reading and the thermometer reading of liquid in sample space is presented in Fig.7. The actual data points are presented in circle and the best fit line was drawn for estimating the mapping parameter. Two sensors have very similar gradient but different offset (drift) due to the location difference and connection issue.

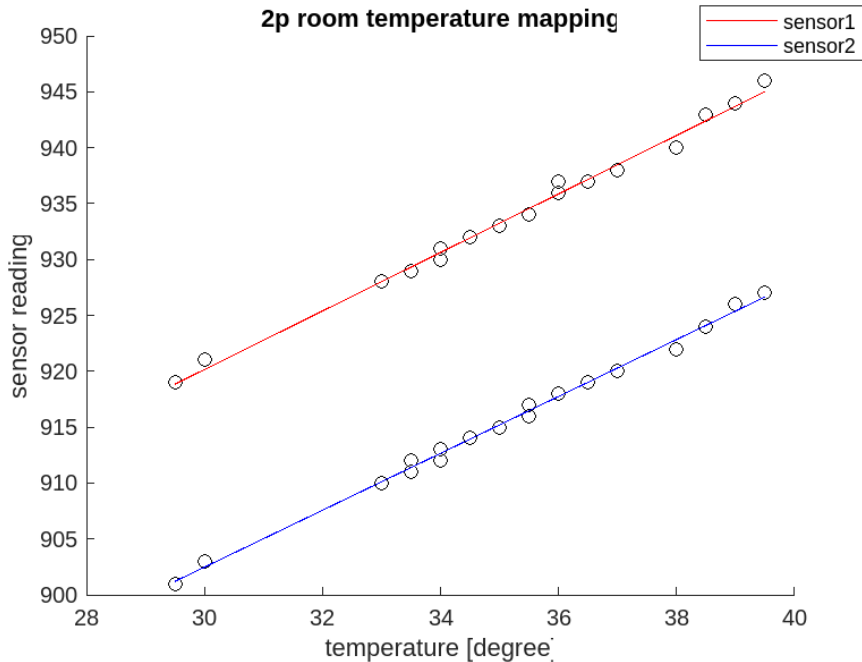


Figure 7: Temperature mapping between sensor locations and the liquid temperature at the sample inner space

The step response is shown in Fig.8. The sensor 2 setpoint was adjusted for step response generation with the actual settle temperature of liquid changed from 34°C to 37.5°C. The controller only used the error calculated from sensor 2, and the overlap of the thermometer recording and the sensor recording are shown in Fig.8b.

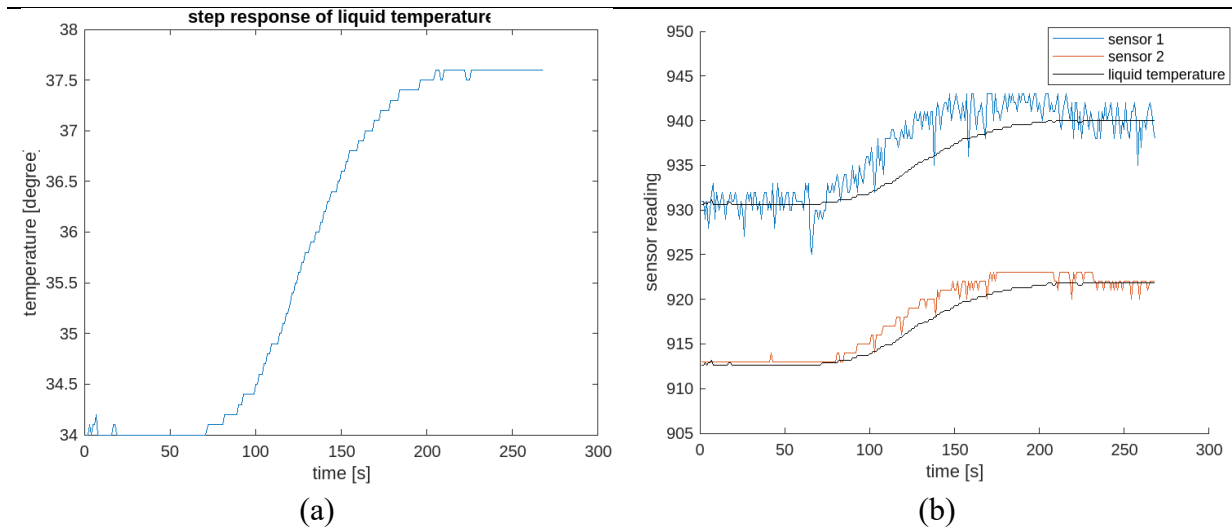


Figure 8: (a) the step response of the liquid temperature, recorded by MM2000. (b) the step response of the two sensor readings.

Within the optimal body temperature range, the control performance of the one sensor proportional control algorithm is shown in Fig.9. In the 15 minutes settlement time, the temperature was kept in $\pm 0.3^{\circ}\text{C}$ range to the averaged stabilised temperature, and the settle temperature changed from 35.7°C to 37.6°C when increase the setpoint from 920 to 925. Thus, the averaged resolution for controlling is 0.4°C per digital numerical value.

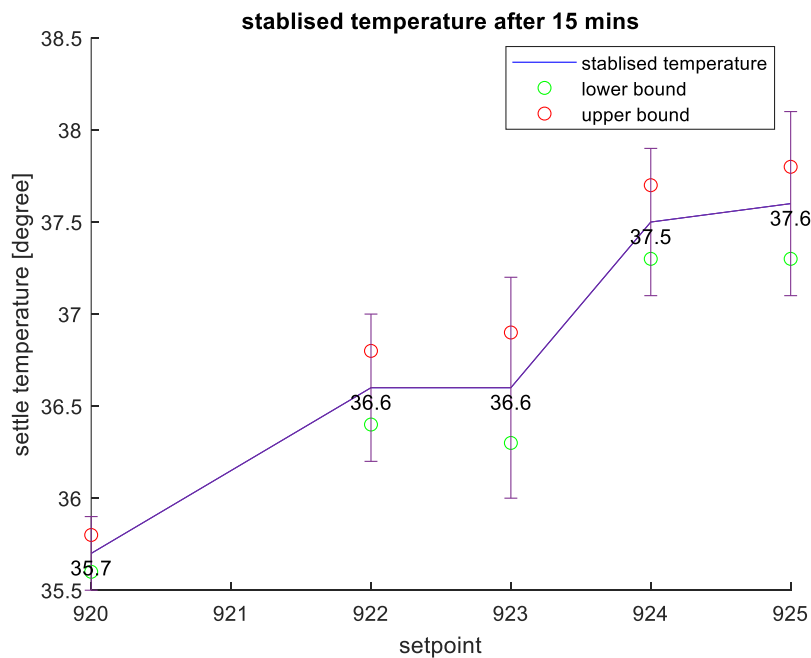


Figure 9: Stabilisation performance of setpoint between 920 and 925, each test last for 15 minutes after the chamber heated up to the set temperature, and the maximum and minimum temperature readings were recorded and plotted.

To visualise the preprocessing of the circuit in Fig.3, the step response (Fig.8) is amplified to use more bits in the ADC. However, the fluctuation of the sensor reading is also magnified.

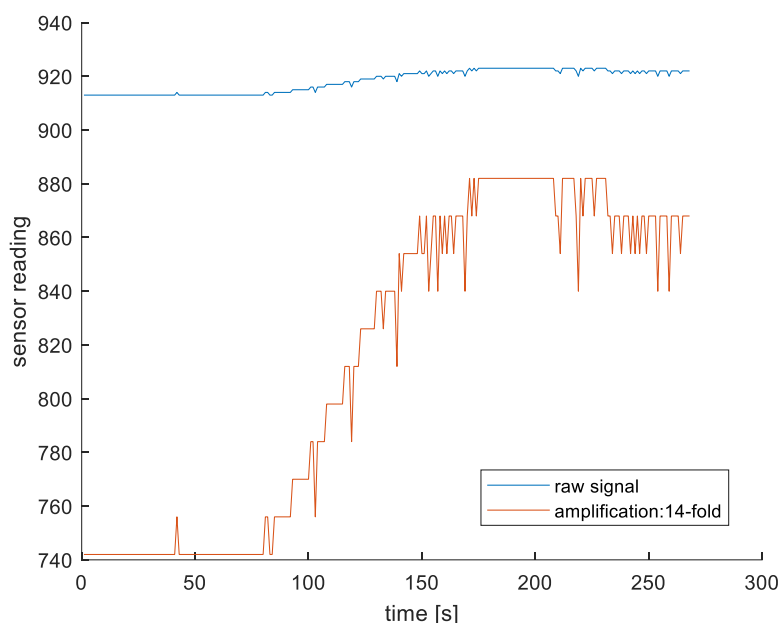


Figure 10: Pre-processed sensor 2 step response signal and the original signal in a same plot for comparison.

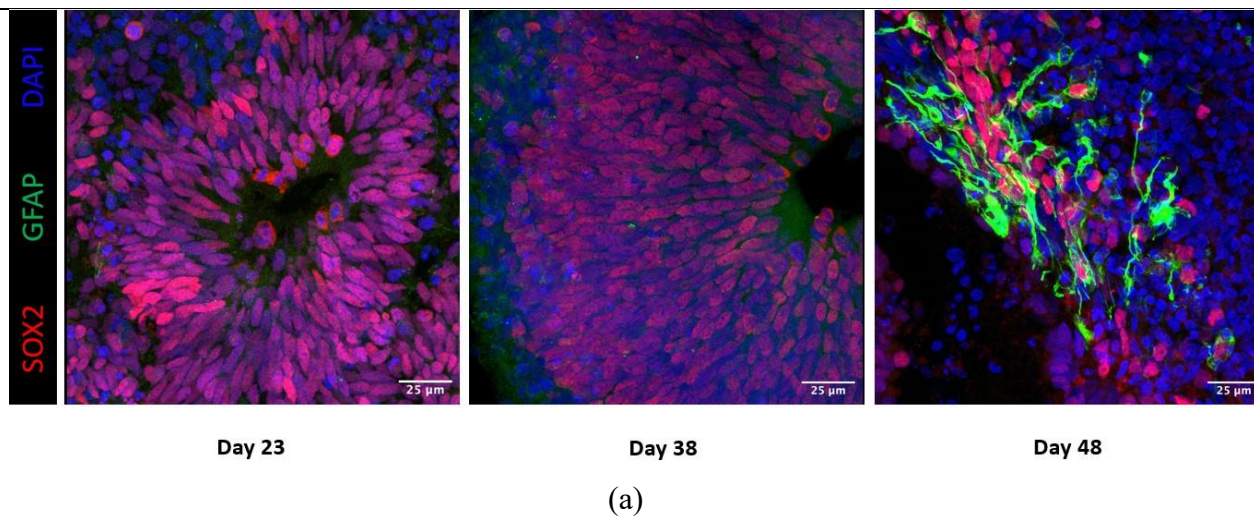
4. Discussion

The system specifically designed with the organoid imaging chamber is able to stably control the temperature around the body temperature. The whole system kept the organoid sterilised on average only one organoid got contaminated after the imaging session among 10 times imaging work. However, there are a few improvements that could be made on this prototype to make it more reliable and secure under this specific operation environment.

The monitoring reading faces the challenge of fluctuation and drifting in the mapping profile due to the environment temperature change, the mechanical connection and the wear of components. This causes problems in the mapping reliability as the data across different days would provide a different mapping profile. There are two possible solutions to this problem. In engineering, a bandgap voltage reference could be added to provide a stable and accurate reference voltage to the circuit. The sensors are able to be recalibrated with this accurate voltage reference which is relatively stable in operation environment variation such as temperature change. For fault alarm setup of the extreme temperature detection, the second sensor works as a reference, and also, to set up a real-time temperature monitoring experiment before imaging with the thermometer. The best checking location of the thermometer probe would be an extra sensor hole in the chamber because the temperature here is less affected by the ambient temperature and would present a similar temperature profile to the connected sensor. During the imaging, exceeding the denaturation temperature could cause irreversible damage to the tissue sample. A protection algorithm is required to switch off the

output power. Furthermore, the setpoint adjustment could be improved by adding a pushbutton potentiometer. After the resolution calibration of the ADC value and the resistance range of the potentiometer, fine adjustments are performable with knob rotation. The input signal quality hugely affects the control outcome. Therefore, the signal pre-processing plays an important role in maintain high signal integrity and signal-to-noise ratio (SNR). One way to keep separable power supply voltages to the integrated circuits (ICs) is to add decoupling capacitors across the op amps power supply. This would also reduce the noise and minimise the cross effect in op-amps.

Both old (5 months old at the time of imaging) samples from our collaborated lab and a new batch (generated in June 2023) in our lab (Fig.11) were observed with 2p microscopy. The investigation aims to validate any temperature impact in the cerebral organoids model and compare the significance of temperature across different ages, maturation and batches of cerebral organoids. Only the shape outline of the samples was captured at the end of the project but no obvious neurons or grouped networks were recorded (Kirwan et al., 2015). The biggest constraint was the imaging instrumentation (artifacts caused by photomultiplier tube power problem) However, the calcium imaging quality could be affected by many factors such as calcium indicators injected into the sample, the spectra, and quantitative signal expression (Grienberger and Konnerth, 2012). The neural progenitor and neurons are presented in the new generated organoids (Fig 9b). The neurons are dominated at the organoid boundary and little neuron-positive signal was expressed compared to the gross cell population, which is likely to be a cause of weak calcium signal expression. A organoid generation technology improvement, vertical mixing with bioreactors which regulate the fluid dynamics (Suong et al., 2021), provided evidence that the self-organisation morphology could inverted by modifying the culture method. The generation and promotion of inhibitory and excitatory neurons could then be varied for matured neural network dynamics study and disease modelling.



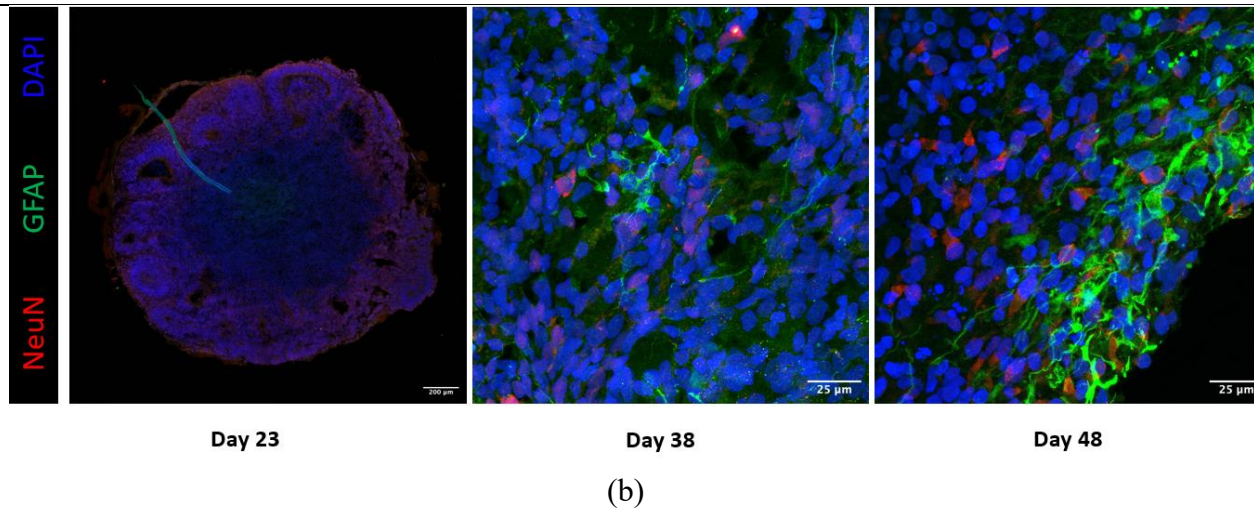


Figure 11: Confocal imaging of the organoids generated in the lab (by Dilara Dupont and Julia Escarrabillilla), where astrocytes is marked by GFAP, (a) neural progenitor marked by SOX2 (b) and mature neurons marked by NeuN.

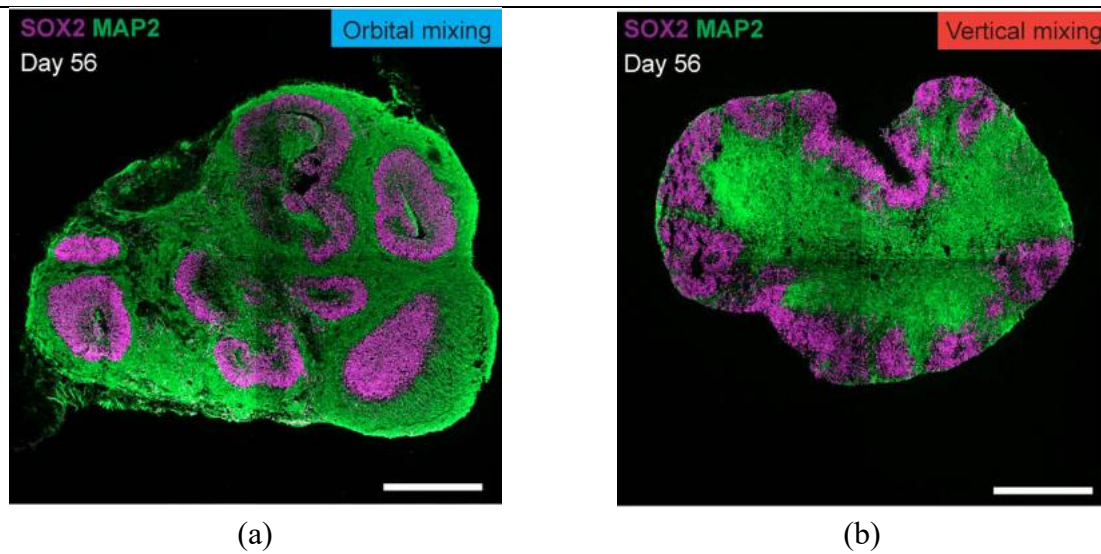


Figure 12: Images from (Suong et al., 2021), with SOX2 marks the neural progenitor cells and MAP2 marks the neuronal layer. (a) with classical orbital mixing, the MAP2-positive neurons is distributed at the edges, (b) whereas with vertical mixing, the neuron-positive area is larger and the SOX2-positive area lays in the peripheral region.

An *in vivo* experiment with a rodent model built a similar temperature control system to manipulate the temperature of specific brain regions through a heat conductive probe (Monteiro et al., 2023). The result showed replicable modulation of the spatiotemporal firing rate profile. Furthermore, spatially, the temperature on the baseline level of neural activity exhibited a nonmonotonic pattern, whereas temporally, an increase in tissue temperature led to a contraction in neural response, while cooling caused dilation in the neural response (Monteiro et al., 2023). These findings of the temperature influence on spontaneous neural activity provides a basis for validation in the organoid model through comparative analysis. There is also

evidence that organoid model responds to artificial neuronal activity modulation by delivering ions and neurotransmitters (Park et al., 2023). The brain organoids illustrated dynamic changes in neuronal activity under calcium imaging with the bioelectronic delivery of potassium ions and neurotransmitter GABA. The successful implementation of the bioelectronic modulation signifies the organoid model development and expands the potential for controlling brain organoid activity for future study. The proposed validation of the temperature regulation in this project shares the vision with this paper.

5. Conclusion

The temperature control system reached the requirement of precise and stable control after calibration. Enhancements could be made in the engineering aspects of the system for a more reliable temperature regulation of the organoid sample. Improvements such as adding sensor calibration reference, high temperature fault alarm and external setpoint controlling knob would elevate this design, making it a more robust engineering solution.

Although replicating the sophisticated thermoregulation in brain is exceptionally challenging, previous attempts aimed at improving organoid culture methods and optimising environmental conditions have sufficiently increase the applicability of the organoid model. At the same time, they have justify the importance of the modulated factors and the technology together could contribute to a better brain organoid model for fundamental mechanisms understanding and the study of neurodevelopmental disease.

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